

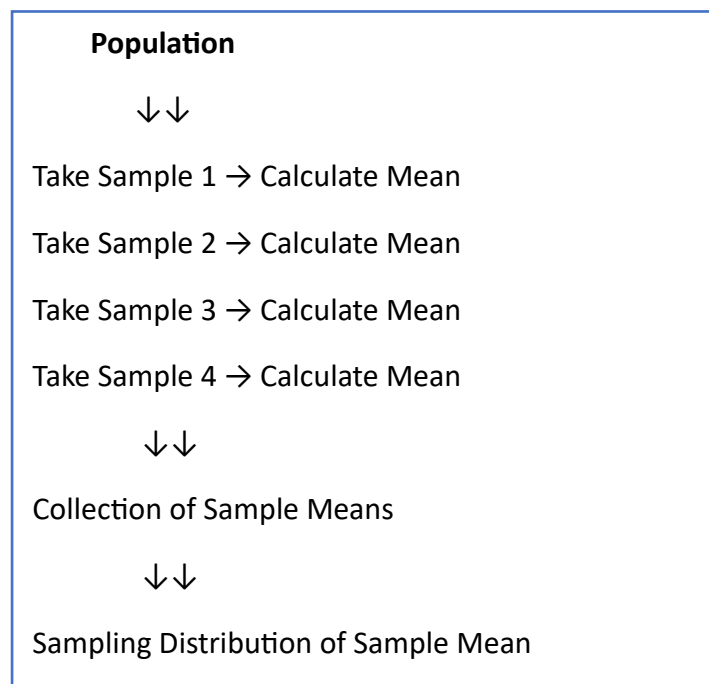
Central Limit Theorem (CLT)

he Central Limit Theorem states that when we repeatedly take random samples of sufficiently large size from a population and calculate the sample mean, the distribution of these sample means becomes approximately normal, regardless of the original population distribution.

sampling distribution

A **sampling distribution** is the probability distribution of a statistic calculated from many samples taken from the same population.

Example



NOTE :

If the **statistic is the mean** -----> we get: Sampling distribution of the **sample mean**

If the **statistic is the proportion** -----> we get: Sampling distribution of **sample proportion**

Sampling Distribution Formula :

$$\mu_{\bar{x}} = \mu$$

The mean of the sampling distribution is : The average of all sample means is approximately equal to the population mean.

Standard Error:

Standard Error

The standard deviation of the sampling distribution is called the **Standard Error (SE)**.

$$SE = \frac{\sigma}{\sqrt{n}}$$

= Standard Error

σ = Population Standard Deviation

n = Sample Size

NOTE:

The Standard Error tells us how much the **sample means vary around the population mean**

$n \uparrow \Rightarrow SE \downarrow$.